

Experimental Evaluation of Multi-Vendor Virtualized RAN using Non-Ideal Fronthaul

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Abstract—In this paper, we present the recent research findings of our experimental investigation into virtualized Radio Access Networks (vRAN) based on non-ideal fronthaul transport. The experiments are conducted in a Telecom Infra Project (TIP) community lab hosted in BT Labs at Adastral Park; BT's global research and development headquarters. A wide range of testing and measurements has been executed to characterise the functional behaviour and performance of vRAN based on a low layer functional split with a focus on using G.fast as the fronthaul link. The performance evaluations reveal that the solution operates well in the presence of transport impairments in excess of typical G.fast requirements, thus proving that G.fast can be a suitable transport option for fronthaul. Testing is carried out on two different multi-vendor systems. The more recent of these systems is built in line with the O-RAN fronthaul specification. It is shown that the solution demonstrates promising throughput performance and is robust to a wide range of fronthaul packet loss rates. What is most significant about these experimental results is that it is real evidence of progress of industry efforts to develop an open, best-of-breed, disaggregated, multi-vendor RAN ecosystem that is envisaged for next-generation RAN. In addition, the fact that non-ideal fronthaul can be supported can significantly improve the business case for a number of deployment scenarios.

I. INTRODUCTION

One of the key trends to support the demanding requirements of the 5G and beyond services is to leverage virtualization into the different parts of the networks. Traditional networks are hardware-oriented, and therefore evaluation of new features and deployment of new standards often require replacement of hardware. As a remedy, softwarisation and cloudification of networks through Software-Defined Networking (SDN) and Network Function Virtualization (NFV) constitute promising solutions to tackle this expensive and time-consuming process, while virtualization also enables a more flexible and reconfigurable network [1]. In particular, the research on virtualized RAN (vRAN) has attracted a great deal of attention [2] [3] in both academia and industry since RAN accounts for a significant proportion of total expenditures of the network operation, e.g., 80% of CAPEX and 60% of OPEX according to the reports in [4] [5].

The next-generation RAN is envisioned to be disaggregated into multiple RAN functions by splitting the baseband processing. With a low layer split (LLS) a base station is formed of a Radio Unit (RU), which remains at the cell site, and a Baseband Unit (BBU), which can be centralised. For vRAN

architectures, the virtualized BBU (vBBU) is run as a software function that can be deployed in virtual machines or containers [6]. vRAN can be implemented with a wide range of functional splits that are discussed by standardization forums (e.g., 3GPP [7] and NGMN [8]); each split offering a different set of trade-offs that can help operators to meet the diverse requirements of future services.

In vRAN, the fronthaul transport link that connects the RU and vBBU needs to fulfil specific requirements (e.g., latency, link capacity, etc.) based on the selected functional split. For the LLS there is often an expectation that ideal transport is required. Transport can be a significant cost, both for CAPEX and OPEX, and the cost of ideal transport may be a significant barrier to deployment for many scenarios in many regions. Hence, it is essential to ensure that vRANs are also capable of operating with non-ideal fronthaul. This can provide more flexibility for operators to use cost-effective transport, which in many cases is already deployed.

There are varying ways to define whether transport (e.g., fronthaul between RU and vBBU, or backhaul between vBBU and core network) is ideal or not. This can be defined as a set of certain thresholds for the expected KPI (e.g., maximum tolerable latency to achieve peak throughput), baselines agreed by project partners (e.g., as defined in [2]), or pre-specified conditions defined by standards, e.g., according to 3GPP 36.932 [9] which says that ideal backhaul transport is characterised by having throughput of up to 10 Gbps and one-way latency of less than $2.5 \mu\text{sec}$, while the non-ideal backhaul is characterised by a wide range of requirements, such as throughput of 10-100 Mbps, and latency of 15-60 msec. According to these definitions, G.fast would typically be considered as non-ideal transport. Other examples of non-ideal transport can include DOCSIS cable, GPON, in-band cellular, ethernet and microwave.

There are several use cases that BT consider for vRAN with non-ideal fronthaul, and therefore characterising the performance of vRAN for such scenarios under various fronthaul link impairments is of great interest. One of the key use cases for BT is based on G.fast fronthaul, which can provide over 1Gbps on short copper lines with a reducing data rate for longer lines [10]. The use case has picocells providing street coverage using G.fast fronthaul, where RUs are deployed on lamp posts and the G.fast fronthaul connections to the vBBU (e.g., at street cabinet) use existing copper infrastructure. This and other potential use cases such as, campus deployments

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with a mix of outdoor/indoor cells using the campus managed ethernet as fronthaul, and temporary coverage (e.g., for special events) using cellular in-band/microwave fronthaul, are detailed in [2].

There are examples of experimental evaluations of low layer functional split solutions [11] and this paper further progresses the industry by demonstrating commercial-grade solutions that are multi-vendor and can function with non-ideal transport. This paper presents key results and vRAN characteristics from the experimental evaluation of two such multi-vendor vRAN solutions. The first solution, which was built against a variation of option 7-2 split, is tested under a wide range of fronthaul link impairments, namely latency, jitter, packet loss, and link capacity. The main test case is focused on G.fast, but a general characterisation is conducted to evaluate the solution for a wider range of transport types. The test results confirm that it is feasible to deploy vRAN with a LLS with G.fast fronthaul transport. The second evaluated solution, allows us to share performance characterization of an early implementation of an O-RAN [12] vRAN based on option 7-2x split. Importantly, this demonstrates interoperability between RU and vBBU provided by different vendors, which is necessary for the establishment of a successful and competitive new ecosystem for next-generation RAN.

The rest of the paper is organized as follows. Section II presents an overview of the functional splits with a focus on LLS, followed by Section III where the experimental setup is introduced along with extensive performance evaluations. Finally, concluding remarks are presented in Section IV.

II. SYSTEM ARCHITECTURE

Fig. 1 depicts the possible functional split options that were considered by 3GPP [7]. This is shown for an LTE protocol stack, but the split locations are equivalent for New Radio. A functional split that is higher in the protocol stack (often referred to as a high layer split), i.e., split options 1-5, centralises part of the baseband processing but most of the processing remains distributed. The industry is mainly gravitating to option 2 for the high layer split. A functional split that is lower in the protocol stack (often referred to as a low layer split (LLS)), i.e., split options 6-8, allows for most of the baseband processing to be centralised. The benefits of centralising more of the baseband includes a better adaptability to non-uniform traffic (by load balancing), a higher user satisfaction (via efficient resource utilization), more effective interference mitigation, a much lower handover time, flexibility in network upgrades, and a lower CAPEX/OPEX [13]. However, these benefits of centralisation come at the cost of strict requirements on fronthaul [14]. Ultimately, operators consider these trade-offs, and choose the split option that best fulfils their service revenue opportunities.

Terminology for disaggregated RAN has varied significantly across industry groups, but some common terminology is now being adopted [8]. Typically we consider the Radio Unit (RU) at a cell site then a distributed unit (DU) and finally a centralised unit (CU). Any baseband processing below the

chosen LLS will go in the RU, while all other baseband processes are in the DU and CU. The chosen high layer split will define which of those processes go into the DU and which go into the CU. The transport link between the RU and DU is typically referred to as fronthaul, whereas the transport link between the DU and CU is typically referred to as midhaul. Our focus is on the LLS, so the DU and CU are combined to form a vBBU, and the fronthaul link connects the RU and vBBU. For clarity, the terminology used for the rest of this paper is RU and vBBU. Interested readers may find the full comparison of these terminologies in Table 3 of [8].

A. Low Layer Splits

When comparing the low layer splits, option 6 can be seen as having the easiest fronthaul requirements, but less of the baseband processing is centralised. Option 8 on the other hand centralises all of the baseband processing, at the expense of significantly tougher transport requirements, which become even more demanding as we consider 5G solutions with wider bandwidths, higher order modulation and massive MIMO. Option 7 is seen as a suitable split for many deployment scenarios, allowing for most of the baseband processing to be centralised, without such extreme transport demands as option 8 (fronthaul throughput scales with user demand for option 7 and when there is traffic it can often still be an order of magnitude more efficient than option 8 [15]). However, the story does not end there. When considering an option 7 split there are actually several variants, which we will discuss here.

The upper part of Fig. 1 illustrates the split options in Physical (PHY) layer for downlink (DL), and the potential locations of the functional splits in an LTE protocol stack as proposed by [7], with a more detailed view of the locations of the option 7 variants. The split option 7-3 which is realized only for DL is the one with the most functions implemented in RU, and split option 7-1 is the most centralised option. For all candidates, the CRC attachment, encoding, and scrambling are performed at the vBBU. For the split option 7-3, all other PHY-layer functions that transform the codewords into frequency-domain IQ symbols are implemented in the RU. When using option 7-2, in the vBBU, the codewords are modulated and mapped into one to several transmission layers. The resultant frequency-domain symbols are conveyed to the RU over the fronthaul link. In the RU, the layered symbols are precoded, and the precoded symbols are mapped to resource elements. Subsequently, the OFDM signal is formed by performing iFFT and appending CP to the IQ symbols [7]. When realizing option 7-1, only iFFT and CP addition is performed in the RU, and the rest of the functions are implemented in the vBBU. For the UL, the locations of the functional splits in an LTE protocol stack are shown in the lower part of Fig. 1 [7]. For all candidates, the FFT processing and CP removal are performed at the RU. For option 7-1, the resource demapping, pre-filtering functions, equalization, and layer demapping are performed at the vBBU, and the rest of the functions reside in the RU [7]. For the option 7-2, only the FFT processing (and CP removal), resource demapping, and pre-filtering are

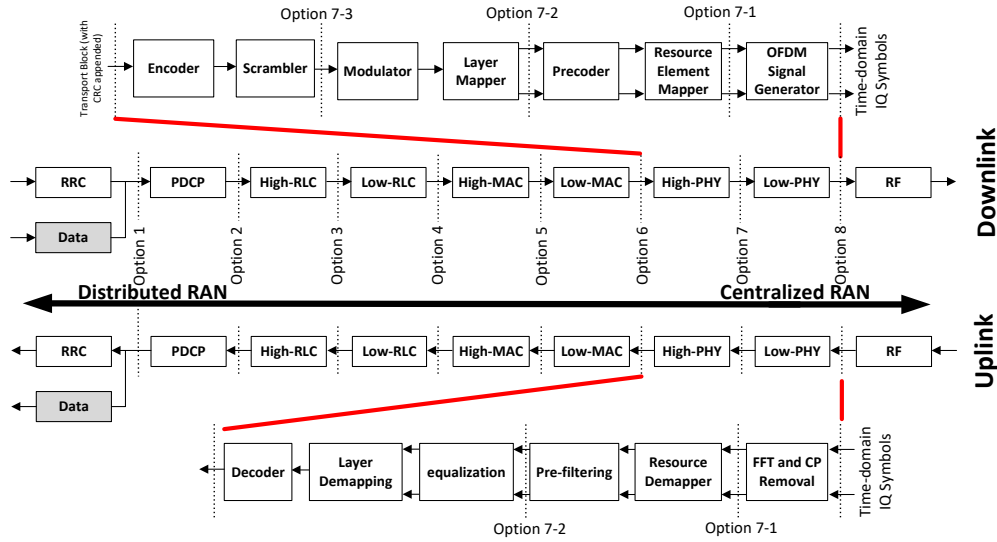


Fig. 1: Functional split options, location of splits, and PHY-layer functions for LLS options as proposed in [7].

performed in the RU, and the rest of the processing is realized in the vBBU.

The O-RAN Alliance has published a specification for the LLS which they describe as being an option 7-2x. This actually allows for a few variations, which could be considered as being close to 7-3 and 7-1 in some cases [15]. First of all there are two categories of O-RAN RU; category A has no support for precoding in the O-RAN RU (relying on precoding in the O-RAN DU) so might be considered to be an option 7-1 split, whereas category B supports precoding in the O-RAN RU, so might be considered closer to an option 7-2 split. There are a number of compression options in O-RAN, one of which is modulation compression. Modulation compression sends data bits to reference a constellation point rather than frequency domain IQ samples, thus being more like a option 7-3 split for downlink.

Both systems tested in this paper use an option 7 split. The first solution was developed prior to the release of the O-RAN low layer split specification, but was built against a high level design that is similar to an O-RAN category B device that uses modulation compression, along with a range of features to allow for resilience to fronthaul impairments. We refer to this as the “pre-O-RAN 7-2 solution”. The second solution was built during the development and release of the O-RAN LLS specification. It is an early embodiment, so some O-RAN features, such as compression, are yet to be implemented. We refer to this as the “O-RAN 7-2x solution”.

B. G.fast as Fronthaul

Fibre-To-The distribution point (FTTdp) networks can complement the roll out of the Fibre-To-The-Home (FTTH) networks in scenarios where deployment of fibre to the end point (e.g., building) is not possible. Currently, there exists a great deal of interest in optimizing FTTdp networks to maximize capacity and deliver high-speed fixed broadband services [16]. The roll out of FTTdp based on G.fast technology [17] has

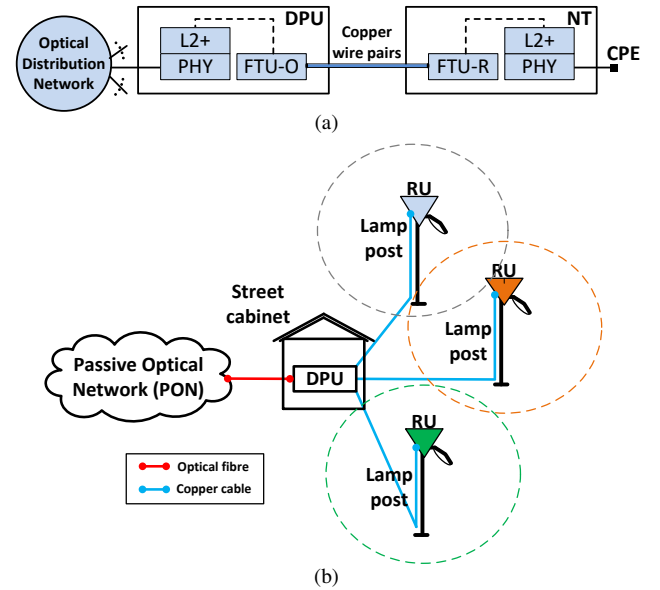


Fig. 2: (a) Reference system model for G.fast [17] (b) Deployment architecture for using G.fast as fronthaul to provide street coverage.

attracted a lot of attention due to its promising performance compared with the related technologies (e.g., VDSL) for short copper loops [18], [19]. The transceivers of the FTTdp networks based on G.fast are denoted by Fast-Transceiver-Units (FTUs). As depicted in Fig. 2a, the FTU at the Optical network segment (FTU-O) and the FTU at the Remote site (FTU-R) are residing in the Distribution Point Unit (DPU) and Network Termination (NT) point, respectively, and are connected via a copper pair. For example, the broadband services delivered to the DPU (e.g., via a Passive Optical Network (PON)) can be distributed among NTs at various customer premises, each of which then provide the service to various broadband applications. One of the many interesting G.fast deployment scenarios is illustrated in Fig. 2b. This can be rolled out based on FTTdp architectures with multi-port

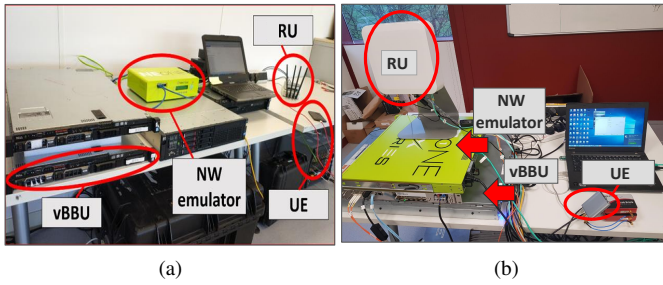


Fig. 3: Testbed setup for (a) pre-O-RAN 7-2 solution (from [14], Note: an upgraded commercial RU was used in testing reported in this paper) and (b) O-RAN 7-2x solution. In each case, the fronthaul connection between the RU and vBBU uses a network emulator to produce fronthaul link impairments.

DPU (e.g., one DPU in a street cabinet connected to several NTs, each NT deployed on a lamp post to provide street coverage). In this, and other G.fast deployment scenarios, the fronthaul transport between DPU and NT must be able to cope with link impairments.

In BT Labs, we want to make sure that, as a part of migration to vRAN-based access networks, the existing copper infrastructure can be utilised as fronthaul without the immediate need for fibre to each cell site. In addition, we want to explore the possibility of various business cases using G.fast as a fronthaul. To this end, we experimentally evaluate the characteristics of vRAN with fronthaul transport that replicates G.fast link characteristics, and also conduct an extensive range of testing to explore characteristics of the vRAN based on low layer functional splits.

III. MEASUREMENTS AND DISCUSSIONS

In this section, we present the experimental results that have been collected after running a wide range of test cases. First we demonstrate that, when we have ideal fronthaul conditions, we can achieve peak capacity. This is also essential for demonstrating that a multi-vendor disaggregated RAN system has successfully integrated a full radio protocol stack that can achieve peak radio performance. After this baseline testing, we then test the two systems under various types of fronthaul impairments, to characterise how the system performance is impacted. The two systems under test, shown in Fig. 3, have very similar configurations. They both implement an LTE radio with a 20MHz FDD carrier that offers 2×2 MIMO in the downlink. For testing with a single UE, a commercial UE was used with the radio transmission taking place within a shielded box. For tests involving multiple UEs, a UE emulator was used in a cabled environment. For link adaptation tests variable attenuators are used between the antennas and antenna ports. The fronthaul impairments are added by a network emulator, which allows for various impairments (i.e., throughput restrictions, latency, jitter, packet losses) to be easily modified. Jitter refers to the packet delay variance that is chosen as a random value from zero to a specified maximum value (denoted by J_{max}) according to a uniform distribution. The fronthaul capacity (B_f) can be limited in both downlink and uplink directions independently, however we always keep it

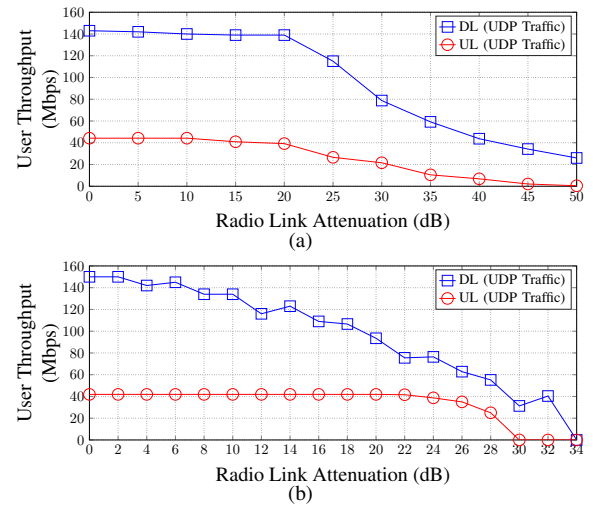


Fig. 4: Throughput performance of DL & UL UDP traffic versus radio link attenuation for: (a) pre-O-RAN 7-2 solution (b) O-RAN 7-2x solution.

symmetrical for these tests. The final fronthaul impairment we experiment with is the packet loss rate (P_L) which is selected as a percentage.

A. vRAN with Ideal Fronthaul

For tests with ideal fronthaul, no fronthaul impairments are added by the network emulator. The expected peak performance for both systems, with their current radio configurations is approximately 145 Mbps in DL and just over 40 Mbps in UL. Fig. 4a and 4b show link adaptation tests for the pre-O-RAN 7-2 solution and the O-RAN 7-2x solution, respectively. Both sets of results confirm a key characteristic that when the signal is strong, peak radio capacity can be achieved in both DL and UL.

Both solutions also demonstrate link adaptation, showing that as the RF signal strength gradually reduces, the radio capacity also gradually goes down. In both examples a user can maintain service with a high attenuation. The O-RAN 7-2x solution is a very early implementation, and it is anticipated that further tuning could be made to keep a user connected with an even higher attenuation. Note that a like-for-like comparison of these two sets of results is not meaningful for several reasons including different transmit powers and the use of different UEs for these tests. However, these tests do demonstrate a baseline performance when fronthaul is ideal.

B. vRAN with Non-Ideal Fronthaul - Pre-O-RAN 7-2 Solution

In order to characterise the behaviour of the vRAN solution, we test the system with each type of fronthaul impairment, one at a time. Fig. 5 shows results for independent tests with fronthaul impairments: latency, capacity, jitter and packet losses.

The first fronthaul impairment we discuss is latency. For many LLS implementations there is an expectation that latency has to be in the order of hundreds of microseconds. Various techniques, such as those described in [14] and Appendix L of [12] can allow the LLS to operate with significantly

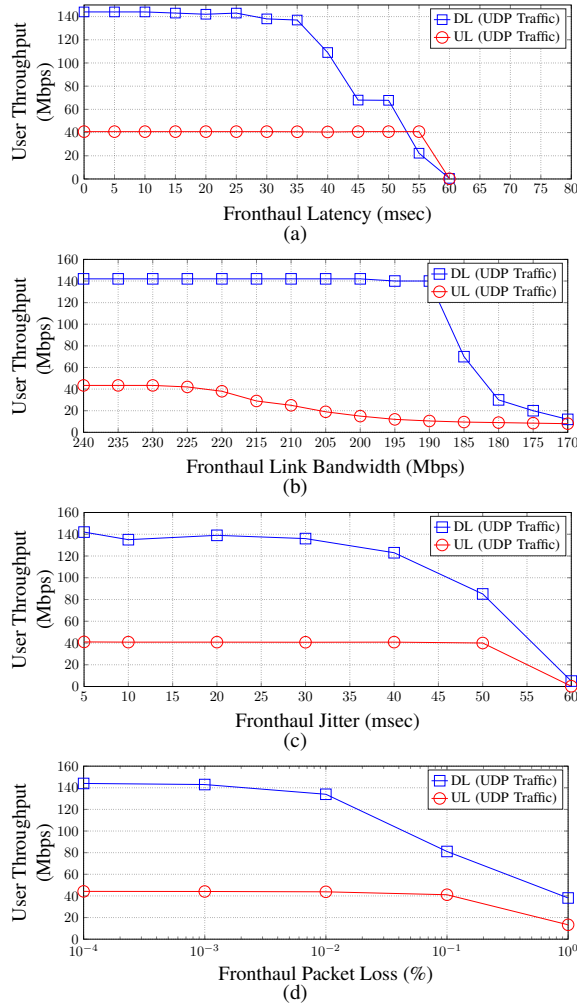


Fig. 5: Throughput characteristics of pre-O-RAN 7-2 solution due to fronthaul impairments: (a) fronthaul latency (b) fronthaul capacity (c) fronthaul jitter (d) fronthaul packet losses.

higher latencies. What we aim to demonstrate here, is to what extent our system under test can tolerate increased latency with the cell remaining operational, and how radio performance is impacted. Fig. 5a presents the DL and UL throughput versus the fronthaul latency with UDP traffic. For these tests no restrictions such as capacity, jitter or packet losses have been applied to the fronthaul. As can be seen, the DL throughput gradually decreases as the fronthaul latency L is increased, while the deterioration of UL throughput is relatively sharp, i.e., for DL throughput to be dropped to zero from its maximum peak radio throughput $T_{DL,max} = 144$ Mbps, the fronthaul latency should be doubled; however, for the UL, only a 10% increase in fronthaul latency results in a throughput drop from maximum radio throughput $T_{UL,max} = 40$ Mbps to zero.

Throughput restrictions on fronthaul are another key concern for the LLS. Two key characteristics to focus on are: the amount of fronthaul overhead relative to the radio capacity, and the rate at which radio capacity degrades when the fronthaul capacity (B_f) becomes constrained. In Fig. 5b the DL/UL

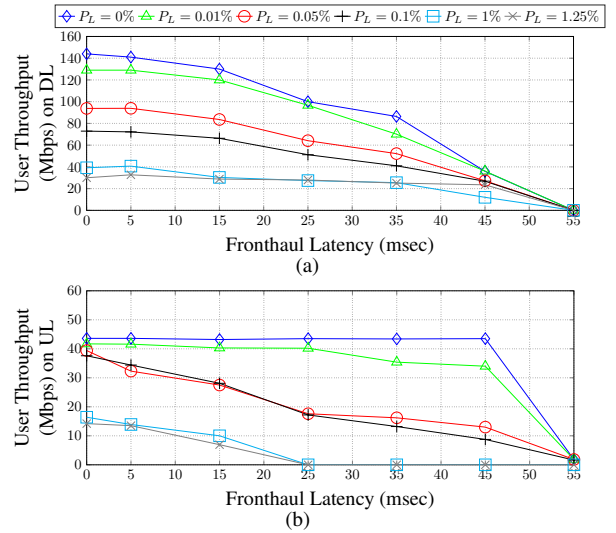


Fig. 6: Throughput performance of UDP traffic versus latency for various packet loss rates (P_L) for pre-O-RAN 7-2 solution.

throughput performance is illustrated as a function of B_f for UDP traffic. When B_f becomes restricted (below 190 Mbps for DL and below 230 Mbps for UL) we see a reasonably steep fall where reducing B_f by 10% for DL results in radio throughput dropping by 90% from $T_{DL,max}$, and for the UL throughput, the degradation has a smoother trend when B_f is gradually restricted. The fronthaul overhead is approximately 30% in the DL before fronthaul becomes constrained, after which point overheads increase. This is good for an option 7 split although additional compression might still be possible. For the UL the overhead is quite significant as compression has not been implemented. Depending on the deployment scenario and the fronthaul being used, an operator may require that compression is implemented in the UL.

Fig. 5c depicts the UDP DL/UL throughput as a function of fronthaul link jitter, J_{max} . For this test, the latency is chosen to be $L = 2$ msec, and therefore the total fronthaul latency introduced to the traffic flow will be chosen from $[2, 2 + J_{max}]$ randomly with a uniform distribution. It can be observed that the DL throughput starts decreasing from $T_{DL,max}$ when the fronthaul jitter is increased to above 30 msec, and reaches zero when J_{max} is as high as 60 msec. However, for the UL, there is no performance degradation even when the J_{max} is as high as 50 msec, but it is abruptly dropped to zero for $J_{max} = 60$ msec.

Fig. 5d presents the UDP throughput performance of DL/UL for various packet loss rates, P_L . It is shown that both DL and UL have peak performance for $P_L = 0.001\%$, which is dropped by 70% for $P_L = 1\%$.

Fig. 6 depicts the UDP throughput as a function of latency for various packet loss rates. In this case, $J_{max} = 10$ msec is used. It is shown that both DL and UL have high resilience to operate in a wide range of fronthaul latencies. However, for $P_L \geq 1\%$, the performance of both DL and UL are highly degraded even for low-latency link conditions.

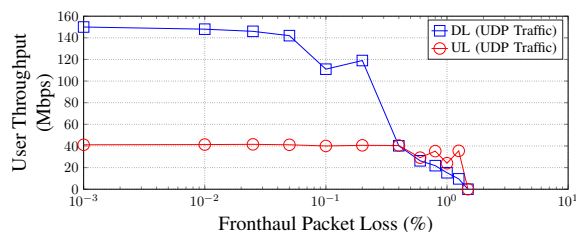


Fig. 7: Throughput performance of O-RAN 7-2x solution in DL and UL against fronthaul packet loss rate (P_L).

C. vRAN with Non-Ideal Fronthaul - O-RAN 7-2x Solution

This section summarises the main performance characteristics from the testing of the O-RAN 7-2x solution. The test setup is depicted in Fig. 3b. Similarly to the previous test setup, a network emulator is placed between the RU and vBBU to emulate various fronthaul conditions. For this set of experiments, we have used UDP traffic, and a limited range of fronthaul impairments are tested. Fig. 7 shows the throughput performance of DL and UL for a wide range of packet loss rates, P_L . For the DL, the throughput starts degrading for P_L on the order of 0.01%, and drops to 40 Mbps for $P_L = 0.4\%$. For the UL, the system operates with the peak performance of 40 Mbps for a wide range of P_L . However, although UL presents a higher tolerance to packet loss compared to the DL (specially for $P_L \leq 0.4\%$), the system does not provide service on neither of UL and DL for $P_L \geq 1.49\%$. This solution is a very early implementation of O-RAN 7-2x, so many optimisations can be expected in the future. In particular, no fronthaul compression has yet been implemented. For this solution to work over a wide range of scenarios with non-ideal transport, it is essential that such features are implemented. Appendix A of [12] provides some examples of compression techniques that can be used for option 7-2x splits, while examples of other features for dealing with non-ideal transport are in Appendix L of [12] and also in [14].

IV. CONCLUSION

Results presented in this paper characterise the behaviours of virtualized RAN in deployments with non-ideal fronthaul. We have shown that it is possible to implement a low layer split (LLS), which offers many potential benefits including better RAN coordination and reduced CAPEX and OPEX, without the need for ideal fronthaul. The characterisation of the solution also highlights the RAN performance impact due to various fronthaul impairments, both individually and as combinations. The exciting reality is that a vendor can offer their vRAN solutions in any region, as a common solution can adapt to any form of transport, ideal or non-ideal. For operators, vRAN deployments can exploit their existing transport assets and each RAN component can be supplied by a number of vendors. This competitive ecosystem will drive towards the future RAN enhancements required for the diversity of next-generation RAN use cases. In future work, we plan to investigate other aspects of the LLS, such as the coordination benefit. As well as the LLS being an open

interface, other RAN interfaces must also be open if we are too ensure the solution can be multi-vendor (e.g., for a vendor-agnostic RAN controller) and so our attention also turns to testing full RAN systems with open interfaces, and ensuring that this emerging ecosystem leads to commercial deployments of multi-vendor disaggregated RAN.

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